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BharatBhasaNet-A unified framework to identify Indian code mix Languages

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ABSTRACT In the rapidly globalizing digital communication sphere, the imperative for advanced multilingual text recognition and identification is increasingly evident. Contrasting the previous works, which were predominantly constrained to 2-3 languages, this paper explores the rich linguistic diversity of India, addressing challenges in automated language processing for 12 languages. BharatBhasaNet, our comprehensive Language Identification (LID) framework, integrates an extensive dataset covering these 12 Indian languages in both native-script and romanized forms, derived from INDICCORP [30], Bhasha-Abhijnaanam [20], and Aksharantar [32] datasets by AI4Bharat[a]. The framework accommodates two models, Roberta-native and Roberta-Romanized, based on attention mechanism and transformer architecture. With its exceptional accuracy of 99.54% in native script and 60.90% in Romanized text, BharatBhasaNet significantly advances language identification, providing broader language coverage than existing LIDs. It excels in interpreting code-mixed sentences , unveiling crucial accuracy patterns related to sentence length, word span, and complexity in multilingual contexts. The framework underwent rigorous testing using a real-time dataset from the National Informatics Center (NIC), achieving an accuracy rate of 92.67%. Overcoming challenges like limited training data and distinguishing similar languages, BharatBhasaNet marks a significant leap in Romanized text identification within diverse linguistic landscapes.

[a]https://ai4bharat.iitm.ac.in

INDEX TERMS Multilingual, Indian, Native, Romanized, Transformer, Attention, XLM-Roberta.

I. INTRODUCTION

LANGUAGE. Identification (LID)plays a critical role in the expansive field of computational linguistics, essential for interpreting multilingual texts [5], [6]. This task assumes greater significance in today's interconnected global environment, particularly in linguistically diverse regions such as India, with its array of languages, scripts, and dialects [2], [3].

Our study focuses on the challenges of multilingual text processing in India, addressing both native [7] and Romanized [8] languages. Prior research has explored various methodologies, from traditional rule-based systems [9] to modern machine learning techniques [10]. However, a substantial gap remains in the subtle recognition of languages, particularly in the detailed analysis of character and word level in code-mixed texts.

To bridge these gaps, BharatBhasaNet is introduced as a pioneering framework in the LID domain. Unlike previous LID systems that are typically capable of managing only 2-3 languages, BharatBhasaNet is uniquely equipped to process 12 Indian languages in native and Romanized scripts. A standout feature of BharatBhasaNet is its proficiency in handling code-mixed sentences, blending native and Romanized languages. It's the first in its field to provide detailed analysis at both the character and word levels, enabling a deeper understanding and more accurate identification of multilingual text

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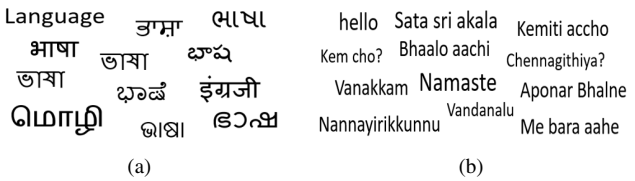
complexities.

BharatBhasaNet marks a milestone in LID technology. As digital technologies, such as translation, ASR, and conversational interfaces, grow in importance [10], [12], [13], BharatBhasaNet emerges as a vital tool in creating resources for less commonly spoken languages [11]. It effectively navigates challenges such as web data noise, sparse datasets [12], and similarities among high-resource languages [14]. Existing LID tools, though valuable, fall short in the intricate and diverse Indian language landscape, particularly in the nuanced handling of code-mixed sentences.

Empowered by advanced machine learning models like Transformer Networks and the Roberta model [18]–[20], BharatBhasaNet excels in identifying complex linguistic patterns in Indian languages [19]. The Transformer networks' capability to capture long-range dependencies [21] and the contextual sophistication of the Roberta model [18], [22] are instrumental in BharatBhasaNet's success. Its strength lies in its unparalleled proficiency in dissecting complex code-mixed sentences at both the character and word levels, setting a new standard in LID.

Native Indian Language identification: In this multi-class classification task focusing on identifying 12 distinct native Indian languages from provided text samples, each language is assigned a specific label (e.g., Bengali as 1, Assamese as 2, up to Punjabi as 12). The primary goal is to accurately categorize input text into one of these 12 categories, effectively recognizing the language it represents. For instance, the successful classification of the input text 'Hello, I am well' as 'English' showcases the model's ability to identify English among 12 native Indian languages. The Figure 1(a) displays examples of Native Languages.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enhances our understanding of the intricacies of language variation and expression within diverse linguistic contexts.

3) **Extensive experimentation and verification**: Rigorous experimentation and validation procedures were meticulously carried out to thoroughly test and affirm the efficiency and reliability of our proposed approach. The framework was rigorously tested on a real-time dataset. The extensive experimentation conducted serves as a testament to the effectiveness and robustness of our methodology in accurately identifying and processing multilingual text, thereby validating its practical utility.

The rest of this paper can be sorted as follows: Section II discusses the related woodworker Multilingual text recognition for 12 Indian languages. III of this paper talks about the data set and experiments. IV discusses the methodology of the proposed pipeline. V presents the experiments and discussion. VI discusses the errors in identifying code-mixed languages. Limitations of the study are given in VII The paper concluded with a detailed conclusion VIII of the paper and future scope.

II. RELATED WORK

In the initial stages of language identification tasks, a binary feature technique based on word and character co-occurrences was utilized to identify 25 Latin alphabet-based languages [23]. Subsequently, an n-gram model [23] was introduced, which focused on sequences of characters to classify different languages in a document. Previous advancements in the code mixed language identification field have shown progress in language identification primarily centered around two languages. Code mixing is of three types [24]. The first is inter-sentential code, where one sentence is of one language, and the other is of another. The second one is intra-sentential code, where many languages are between single sentences, which means there is a switching between languages within the same sentence. The third one is word-level code, where switching happens within words. For English-Bodo code-mixed text classification sourced from Facebook, Decision Trees, Naive Bayes, and Multi-Layer Perceptron models were employed [25].

There was a code mix tag set using an acronym, Universal, Named Entity, Undefined, and Word Mix other than English and Bodo. Language identification with the use of machine learning classifiers was dominant. The author used n-gram and dictionary-based features. However, these methods showed limitations in understanding and addressing multilingual complexities in context. Similarly, in the context of English-Kannada [26] code-mixed sentences, Support Vector Machine, Naive Bayes, and Random Forest classifiers were used, accompanied by TF-IDF and Bag of Words feature extraction methods. Bag of words [27] represented the text unordered without seeing their grammar. In a bag of words, all the unique words are put together and make the vocabulary, and each sentence or document will get a unique vector in terms of word frequency. Nonetheless, Bag of Words models faced issues related to semantics and out-of-vocabulary terms, partially mitigated by the bi-gram model. Traditional techniques assign equal importance to all words in a sentence or document, lacking discrimination in word relevance. However, TF-IDF [28] addresses this by accentuating the significance of specific words, elevating the importance of those deemed most significant while diminishing the weights of commonly occurring terms through IDF. Nevertheless, despite its efforts to prioritize certain words, the TF-IDF feature extraction method grapples with the challenge of out-of-vocabulary terms.

The Introduction of a Bidirectional Long Short-Term Memory (BLSTM) model [29] for Hindi-English code-mixed data proved insightful, employing word embeddings for English and character-based embeddings for Hindi Romanized words, showing promising outcomes at the word level. For generating these word embeddings, the Word2Vec package was used, followed by the skip-gram technique.1-gram, 3-gram, and 5-gram embedding types were used for the classification of English, Hindi, and Named Entity. The model could capture the semantic meaning of long sequences. However, the applicability of this model was confined to Hindi Romanized and English words. In the context of Malayalam-English code mixed data [20], researchers extensively employed BERT models and fine-tuned these models using a data set comprising 50 thousand code-mixed samples. However, the studies were limited in their scope, lacking thorough comparisons with alternative machine learning or deep learning models. These studies predominantly leaned towards transformer-based methodologies without broader comparative analyses with other model frameworks.

III. DATASET AND EXPERIMENTAL SETUP

For this study, a diverse dataset encompassing a representative sample of Indian languages was curated. For the identification of Indian languages, there is a dataset IndicCorp [30] in the AI4BHARAT research lab introduced by IIT Madras to make research possible in the field of Indian languages. The Dataset comprises text documents spanning multiple domains, including news articles, literature, social media posts, and official documents, to evaluate the proposed methods comprehensively. It covers languages like Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Punjabi, and more, each presenting distinct linguistic characteristics and challenges.

A series of experiments were conducted on the curated Dataset to assess the effectiveness of the proposed approach. The experiments evaluated language recognition accuracy and the model's ability to handle code-switching scenarios where multiple languages are intermingled within a single text.

The RoBERTa model, pre-trained on a large multilingual corpus, was fine-tuned on the Dataset using a cross-entropy loss function. To ensure optimal convergence and performance, dedicated attention was given to configuring the model's hyper-parameters, including learning rate, batch size, and training epochs.

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Traditional machine learning models, including Support Vector Machines (SVMs), were also used to establish a baseline for comparison. These models were trained on engineered features like character n-grams, frequency distributions, and linguistic rules.

The evaluation process involved random sampling of the Dataset into training, validation, and test sets, maintaining language distribution proportions. Precision, recall, F1-score, and accuracy were used as performance metrics, and detailed error analysis was conducted to understand the specific challenges faced by the models, such as misclassification due to phonetic similarities and font variations.

By conducting rigorous experiments and comparisons, this study seeks to elucidate the efficacy of utilizing advanced language models like Roberta in enhancing multilingual text recognition and identification systems, especially within the context of the intricate linguistic landscape presented by Indian languages.

A. NATIVE TRAINING AND TESTING DATASET

The native training and testing set was made from the INDICCORP [30] dataset in AI4bharat. It consists of 12 Indian languages. The Number of training sentences and Number of testing sentences for each Native language are presented in Figure 3.

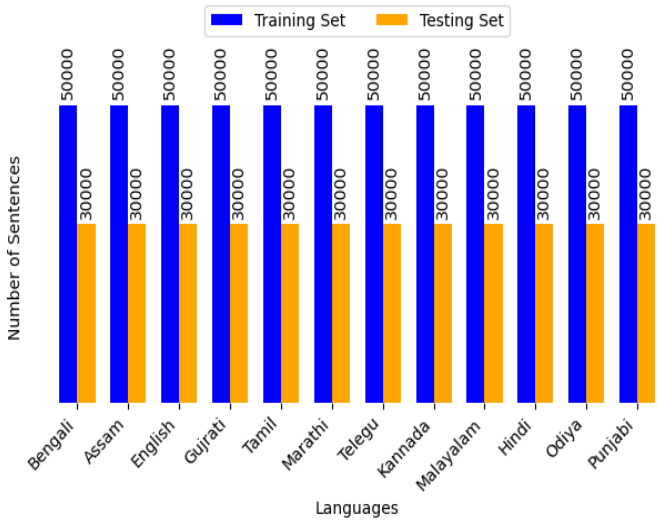

FIGURE 3. Number of sentences in 12 Native Indian Languages, categorized by training (blue) and testing (orange) sets. The dataset contains 50,000 training and 30,000 testing sentences for each language.

Preprocessing: The textual files from the IndicCorp dataset underwent processing, including removing memorable characters, numbers, and punctuation from the sentences. Examples of sample sentences are shown in Figure 4. The Indian native language filtered data was structured as "Sentences" and "Language", each represented by a specific numeric code ranging from 1 to 12. For languages other than English, English characters were excluded. Following this, rows with missing values were removed, and the cleaned Dataset was exported to a CSV file. This meticulous process aimed to clean and organize the textual data, preparing it for training, comprising 600,000 (50,000 sentences per language) lines for training and 360,000 (30,000 sentences per language) for testing.

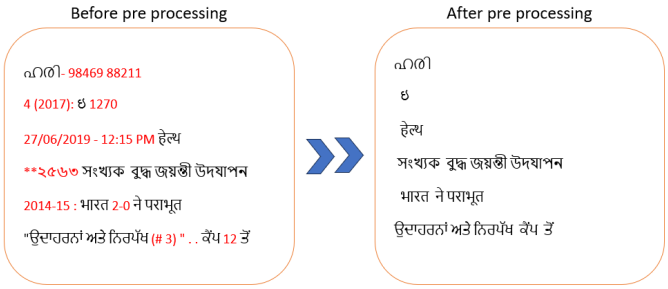

FIGURE 4. Native Indian language sample sentences before and after pre-processing. Red text indicates eliminated numbers and special characters.

B. ROMANIZED TRAINING AND TESTING DATASET

To obtain the romanized data, information was sourced from two datasets: Bhasha-Abhijnaanam [20] and Aksharantar [32] from the AI4Bharat website. These datasets consist of sentences or words originally scripted in various Indian languages, transliterated into the English script, also known as the Romanized script. Details regarding the Number of testing sentences available for each native language are presented in Figure 5.

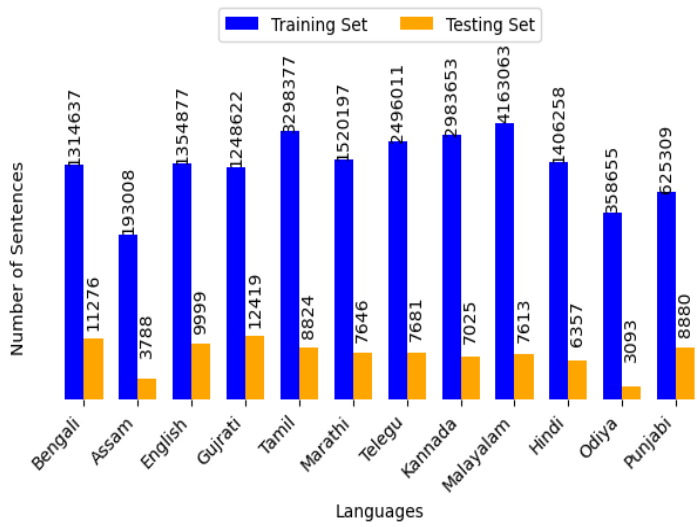

FIGURE 5. Number of sentences in 12 Romanized Indian Languages, categorized by training (blue) and testing (orange) sets.

Preprocessing Romanized Dataset: Data from various languages sourced from Aksharantar (Test and Train files) and the Abhijanam dataset were used for training purposes. Subsequently, this data was processed, including the removal of memorable characters, numbers, and punctuation from the sentences, and amalgamated into a unified file. For testing

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purposes, the Aksharantar (Valid) dataset was exclusively employed. Following this consolidation, the training and testing Data Frames were augmented by introducing "Sentence" columns, paired with corresponding "Language" columns containing singular word values

C. REAL TESTING DATASET

In addition to the datasets sourced from AI4Bharat, a testing dataset provided by the National Informatics Centre (NIC) was also incorporated to evaluate the reliability and suitability model for real-world applications. A significant portion of the dataset is comprised of sentences that are either written entirely in English or solely in native regional scripts. Another subset of the dataset contains mixed-language sentences, which feature a combination of English and regional language elements. Additionally, a portion of the dataset also includes sentences written in a combination of English and Romanized scripts of regional languages. This dataset comprises approximately 1,600,000 sentences, encompassing a diverse range of linguistic compositions.

IV. METHODOLOGY

A. TOKENIZATION

Our study delves into the examination of Roberta, a significant variant of BERT, renowned for its utilization of the Byte Pair Encoding (BPE) technique for subword tokenization. Two widely recognized methods, Word Piece and Byte Pair Encoding, stand out in subword tokenization. Word Piece, as detailed in [33], disassembles words within a sentence into smaller units called word pieces, accompanied by adding special boundary symbols, effectively addressing ambiguities in language. Conversely, Byte Pair Encoding (BPE), as described in [34], dissects words into smaller subword units by replacing frequent byte sequence pairs with previously unused byte sequences. This methodology holds particular advantages, especially for languages like those prevalent in India, characterized by complex morphological structures and agglutinating traits. By segmenting words into subword units, BPE offers a refined approach to a text representation, enabling models like Roberta to gain a more intricate and nuanced understanding of these complicated linguistic structures. In contrast, Sentence Piece models, which function similarly to Word Piece, have gained attention for efficiently handling subword tokenization tasks, particularly in languages with complex morphology and agglutinating features.

B. ATTENTION MECHANISM

The attention mechanism, a cornerstone in transformer model architectures, is pivotal in determining the relevance of words in a sequence. It enables the model to focus on significant words while minimizing the impact of less relevant ones. This functionality is particularly crucial in Natural Language Processing (NLP), where it has been instrumental in enhancing the performance of transformer-based models.

In text classification, especially in models like Roberta, the attention mechanism is employed to extract and analyze semantic nuances within sentences. It identifies individual words' varying roles in conveying the overall semantic meaning. This process is facilitated by three critical components derived from input embeddings: the Query (Q), Key (K), and Value (V) matrices. The Query matrix pinpoints the word under consideration, the Key matrix evaluates its relevance across the input sequence, and the Value matrix contains the information to be extracted based on this determined relevance.

The mechanism initiates with a score calculation, denoted by (1). This step quantifies each key's relevance to the query, essentially deciding how much attention should be given to different parts of the input for a specific word or token. A critical aspect of this process is using a scaling factor related to the dimension of the key vectors. This scaling is essential to prevent minimal gradients during training, a potential issue when passing through the softmax function. It ensures balanced gradients, thereby facilitating stable and practical learning.

Subsequently, a softmax normalization is applied. This step converts the relevance scores into a probability distribution, guaranteeing that they collectively sum up to 1.$\sqrt{d_k}$ A scaling factor used to stabilize the gradients during the training process. It's the square root of the dimensionality of the keys d_k.

The resulting probabilities are then used in the Value matrix(2), culminating in the final output of the attention mechanism.

The attention mechanism's ability to dynamically adjust focus on various parts of the input sequence, as informed by the interplay of the Query, Key, and Value matrices, is critical to its effectiveness. This dynamic focusing is essential for accurately capturing contextual nuances in language, allowing transformer models to excel in complex language processing tasks.

$$\text{Score}(Q, K) = QK^T \tag{1}$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{\text{Score}(Q, K)}{\sqrt{d_k}}\right) V \tag{2}$$

C. TRANSFORMER NETWORKS

Transformers, introduced in the landmark paper "Attention Is All You Need" [35], have revolutionized natural language processing (NLP) by moving away from traditional sequential data processing. Instead, they utilize self-attention mechanisms that allow for parallel processing of input data, adeptly capturing relationships between words regardless of their positions within the text. This architectural innovation shown in Figure 6 comprises two core components: an encoder and a decoder, each consisting of layers incorporating self-attention and feed-forward neural networks.

At the heart of the Transformer's functionality is its self-attention mechanism. This process involves computing the significance of each word relative to others within a sentence, enabling the generation of context-aware embeddings using equations (1) and (2). The encoder and decoder structure of

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the Transformer further enhances this mechanism. Following a specific layer normalization(LayerNorm) and sublayer function (3), each encoder layer includes a multi-head self-attention mechanism and a fully connected feed-forward network. The decoder mirrors this structure with an added layer for multi-head attention over the encoder's output(4), facilitating a complex encoding-decoding process that captures extensive dependencies and contextual information.

A critical aspect of the Transformer is its multi-head attention mechanism (5) (6). $head_1, \ldots, head_h$: These represent individual attention heads, where h is the number of attention heads. Each attention head performs its own attention computation. The output weight matrix W^O, is used to linearly transform the concatenated output of the attention heads to match the desired output dimensionality.W_i^Q, W_i^K, W_i^V : The weight matrices W_i^Q, W_i^K, W_i^V : specific to the ith attention head, used to linearly transform the query, key, and value matrices, respectively.Attention(QW_i^Q, KW_i^K, VW_i^V) : Denotes the attention mechanism applied to the transformed query, key, and value matrices corresponding to the ith head.

This feature enables the model to attend to different segments of the input sequence concurrently, processing information from various representational subspaces. This multi-head approach is instrumental in the Transformer's ability to understand and interpret complex linguistic patterns. In conclusion, the mathematical formulations underlying Transformer networks, especially the self-attention mechanism and the encoder-decoder architecture, are critical to their effectiveness in diverse NLP tasks. Their proficiency in handling tasks like machine translation, text generation, and sentiment analysis is attributed mainly to their capability to manage long-range dependencies and contextual nuances in text sequences.

$$\text{LayerNorm}(x + \text{Sublayer}(x)) \tag{3}$$

$$\text{DecoderLayerOutput} = \text{LayerNorm}(x + \text{MultiHeadAttention}(x, \text{EncoderOutput})) \tag{4}$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{5}$$

$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{6}$$

D. ROBERTA MODEL

Roberta, standing for "A Robustly Optimized BERT Pretraining Approach" [36], is a state-of-the-art language representation model that represents a significant enhancement over its predecessor, BERT (Bidirectional Encoder Representations from Transformers). Developed by Facebook AI, Roberta refines BERT's architecture by undergoing an extensive pre-training process on a substantially larger dataset and for a longer duration. This comprehensive training enables the model to handle diverse languages and tasks, producing deeply contextualized embeddings capable of capturing intricate syntactic and semantic patterns in text. The key to Roberta's enhanced performance lies in this extended pre-training phase, which equips it with a more profound understanding of language structures, making it exceptionally

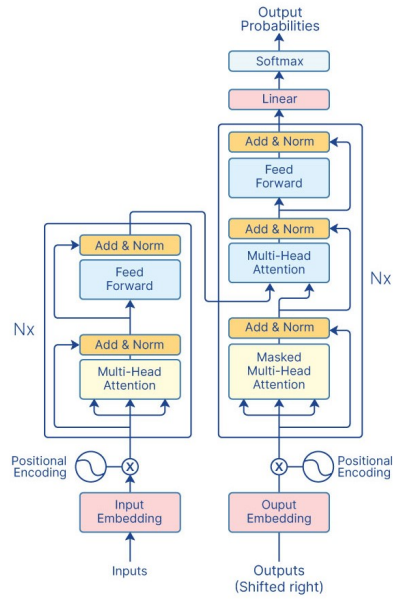

FIGURE 6. The Transformer architecture consisting of the Encoder and Decoder blocks

adept at various natural language processing tasks. Roberta's advancements have cemented its role as a highly effective tool in NLP, excelling in tasks ranging from sentiment analysis to complex question-answering and setting new benchmarks in language processing capabilities.

Further augmenting its capabilities, Roberta's advanced training approach allows for more nuanced and accurate language interpretations, significantly improving task-specific applications. It demonstrates exceptional proficiency in understanding context, subtlety, and the complexities of different languages, which is vital in the ever-expanding realm of NLP. This adaptability and depth of understanding make RoBERTa not just an evolution of BERT but a transformative model that pushes the boundaries of what's possible in language understanding and text analysis. Its success underscores the importance of extensive pre-training in developing models that can effectively navigate the nuances of human language.

E. FINE TUNING XLM ROBERTA

In this section, the fine-tuning of the XLM-RoBERTa model, a prominent pre-trained language model, for the specific application of language detection is detailed. Our objective is to refine a robust and precise model that can effectively differentiate between multiple languages within a text, a task of increasing relevance in the diverse linguistic landscape of today's digital content.

Our model's development, training, and evaluation stages were streamlined by employing the Hugging Face Transformers library, renowned for its efficiency and versatility in NLP tasks. The dataset, a critical component of our research, was derived from a comprehensive CSV file. Before train-

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ing, meticulous pre-processing of this data was addressed, addressing missing values and implementing a robust tokenization process to adapt the textual data for our model's consumption.

A distinctive aspect of our methodology is adopting a custom metric, 'compute metrics,' designed to assess the model's performance with a focus on accuracy and F1-score. This dual-metric approach provides a balanced evaluation, capturing the model's precision and recall capabilities.

The fine-tuning process was conducted over multiple epochs, precisely ten, with a carefully selected adjustable learning rate of 2e-5. This choice of learning rate, along with the optimization of batch sizes to 1280 for training and 2560 for evaluation, was instrumental in achieving the desired model efficiency. Incorporating dynamic padding was another critical strategy, allowing for efficient handling of texts of varying lengths.

Our training approach was further enhanced using mixed-precision training (FP16), significantly accelerating the process by leveraging GPU capabilities. This improvement not only increased training efficiency but also contributed to the scalability of our model training.

F. BEAM SEARCH

In our research, beam search was effectively implemented to extract the most probable language sequence from the word-wise language probability values provided by the RoBERTa model. Beam search, renowned for its efficiency in sequence prediction tasks, particularly excels in scenarios with an extensive output space. This heuristic algorithm distinguishes itself from traditional search methods by limiting the expansion of nodes at each level, a constraint determined by the beam width, denoted as (b). The process commences with the initial expansion of nodes, followed by a scoring phase as in equation (7) based on the language probabilities output by the RoBERTa model. In (7), the score $\text{Score}(Y_{t-1}, y_t)$ represents the likelihood of extending the partial sequence Y_{t-1} with the next token y_t, computed as the sum of the score of Y_{t-1} and the logarithm of the conditional probability $P(y_t|Y_{t-1})$. Crucially, at each level of expansion, the algorithm retains only the top b nodes with the highest scores for further consideration.

This selection process is governed by a specific mathematical formula (8). This process helps to assess the probability of each potential extension to the sequence. In (8), the set Y_t consists of the b most probable partial sequences at time step t, determined by selecting the top b sequences based on their scores according to $\text{Score}(Y_{t-1}, y_t)$. The beam search algorithm efficiently explores the space of possible sequences by considering multiple candidate sequences simultaneously and selecting the most promising ones for further exploration in the decoding process. Such a strategic approach balances computational efficiency and accuracy in predicting language sequences. The algorithm concludes its operation upon meeting a predetermined condition, like reaching the maximum sequence length or encountering an end-of-sequence marker.By integrating beam search with the RoBERTa model output, accurate prediction of language sequences within multilingual texts could be achieved. This methodology significantly boosted the efficacy of our language detection model, demonstrating the potency of combining advanced NLP techniques with optimized search algorithms in addressing intricate language processing tasks.

$$\text{Score}(Y_{t-1}, y_t) = \text{Score}(Y_{t-1}) + \log P(y_t|Y_{t-1}) \tag{7}$$

$$Y_t = \text{top}_b(\text{Score}(Y_{t-1}, y_t)) \tag{8}$$

G. PROPOSED PIPELINE

Two models, Roberta Native, and Roberta Romanized, were introduced, as outlined in Figure 7, to address the challenge of identifying code-mixed sentences comprising multiple languages. Initially, the Roberta Native model classified words in a sentence into categories such as English, Hindi, or any regional language. However, an issue arose where Romanized words within mixed sentences were often declassified as "English." To mitigate this problem, the Roberta Romanized model was introduced. This specialized model focused on reclassifying Romanized words previously labeled as "English." It differentiated between "True English," "Romanized Hindi," and "Regional Romanized" words, subsequently transliterating the latter two categories into separate Hindi and Regional Native Words. The sentence was reconstructed by retrieving the stored indices of Hindi words, Regional native words, transliterated Hindi words, and regional words. The Roberta Native model was then re-applied to classify "True Hindi," "True English," and "True Regional" words. The pipeline constructed language sequences for mixed sentences through a beam search process detailed in Figure 8.

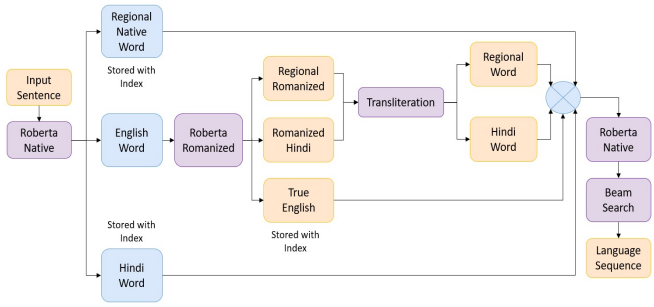

FIGURE 7. BharatBhasaNet efficiently deciphers code-mixed sentences in 12 Indian languages using a streamlined two-model approach. Initially, the Roberta Native model classifies each word as English, Hindi, or a regional language. For Romanized words often mislabeled as English, the Roberta Romanized model intervenes, accurately differentiating between true English and Romanized Indian languages. This is followed by a transliteration of Romanized words into their native scripts and a meticulous reconstruction of the sentence. The process concludes with a final application of the Roberta Native model, ensuring precise language identification, adeptly revealing the complexities of code-mixing in multilingual texts.

V. EXPERIMENT AND DISCUSSION

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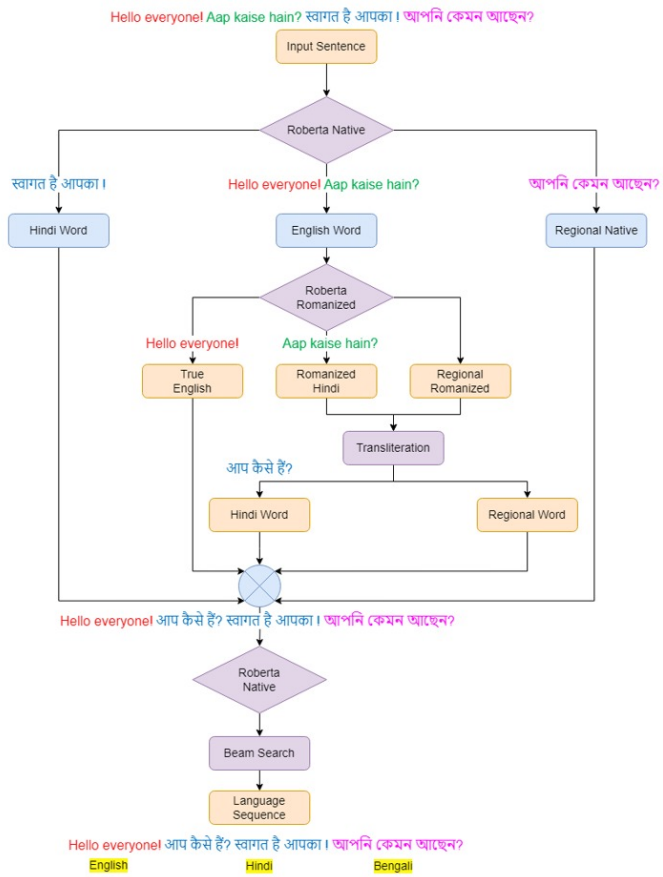

FIGURE 8. Detailed overview of a code-mixed 12 languages identification of a novel pipeline. Input sample with mixed code of Native and Romanized sentences: red (English), green (Hindi Romanized), blue (Hindi native), and pink (Bengali native) highlight specific language components. Output includes the final transliterated sentence in their native language: red (English native), green (Hindi native), and pink (Bengali native) denote respective language components. Language classification tags (yellow) accompany the output, providing insights into the identified code-mixed languages.

A. RESULT ANALYSIS

XLM-Roberta pre-trained model initially trained on 2.5 TB of filtered Common Crawl data spanning 100 languages is utilized. This variant of ROBERTa, known as XLM-ROBERTa, is a multilingual model boasting 125 million parameters, 3072 feed-forward hidden states, 768 hidden states, 12 layers in the transformer blocks, and eight self-attention heads. The Adam optimizer and the GELU (Gaussian Error Linear Unit) activation function were employed for our experiments. All experiments used a CPU - Intel i7 12th Gen @ 2.70GHz, 128 GB RAM, and an NVIDIA 3090 TI graphics card with 24 GB memory. The results obtained from these experiments are detailed in the subsequent sections.

B. XLM ROBERTA VS SVM

The performance comparison between the XLM-Roberta Native Model and SVM Native classifiers is outlined in Table 1. Notably, in the case of native scripts, the XLM-Roberta model demonstrated superior results compared to SVM-based classifiers. To convert textual data into numerical representations for both Native and Romanized cases, feature extraction methods such as Count Vectorizer and TF-IDF Vectorizer were employed. For the SVM classifier in the Native results, the Count Vectorizer feature extraction method was utilized, and a Stratified K-Fold validation approach with k=10 was implemented. Additionally, with the TF-IDF feature extraction, GridSearchCV was employed to select the optimal parameter for tuning

Regarding the Romanized results in both SVM and XLM-Roberta, detailed in Table 2, Count Vectorizer was exclusively employed without using validation approaches.

Similarly, TF-IDF was utilized without GridSearchCV. Notably, the XLM-Roberta model for Romanized text exhibited superior performance across all evaluation metrics compared to the SVM-based approach for Romanized text.

TABLE 1. Performance Comparison of XLM-Roberta Native Model and SVM Native Classifiers with Feature Extraction Methods

Model	Testing data	P	R	F1	Acc
SVM Native+Count Vectorizer	360000	96.93	96.55	96.65	96.54
SVM Native+TF-IDF Vectorizer	360000	97.41	97.35	97.35	97.35
XLM ROBERTA Native	360000	99.55	99.54	99.54	99.54

TABLE 2. Performance Comparison of XLM-Roberta Romanized Model and SVM Romanized Classifiers with Feature Extraction Methods

Model	Testing data	P	R	F1	Acc
SVM Romanized	94600	42.96	21.82	17.78	21.82
XLM ROBERTA Romanized	94600	63.90	60.90	61.31	60.90

1) XLM ROBERTA Native Word Wise confusion analysis

The word-wise confusion matrix for XLM ROBERTa is shown in Figure 9(a), Figure 9(b).In the figure, the 2-word and 3-word confusion matrices are observed. A significant confusion is noted between Bengali-Assamese, Hindi, and Marathi languages. The representation of labels is defined in the form of languages in the table. The accuracy of 2 and 3 words was tested, and after the analysis, individual testing accuracy for Bengali, Marathi, and Hindi in confusion matrices increased. This relationship is presented in Figure 10.

Two-word and three-word confusion matrices were analyzed in the SVM and Count Vectorizer feature extraction method. These matrices illustrate a notable pattern of confusion, particularly between Bengali-Assamese, Hindi, and Marathi languages. However, a different feature extraction TF-IDF feature extraction method is used. In that case, there is a common confusion between Telugu and any Indian Language. However, the confusion percentage decreases for the

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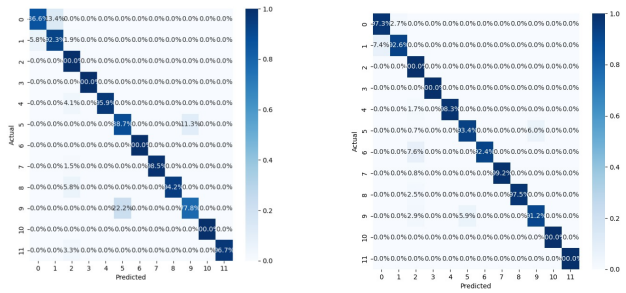
Telugu language and for both feature extraction methods as the words increase, and the confusion percentage decreases for other languages.



(a) (b)

FIGURE 9. Comparision of Confusion matrices for XLM ROBERTA native: (a) 2 words confusion matrix for XLM ROBERTA native, (b) 3 words confusion matrix for XLM ROBERTA native

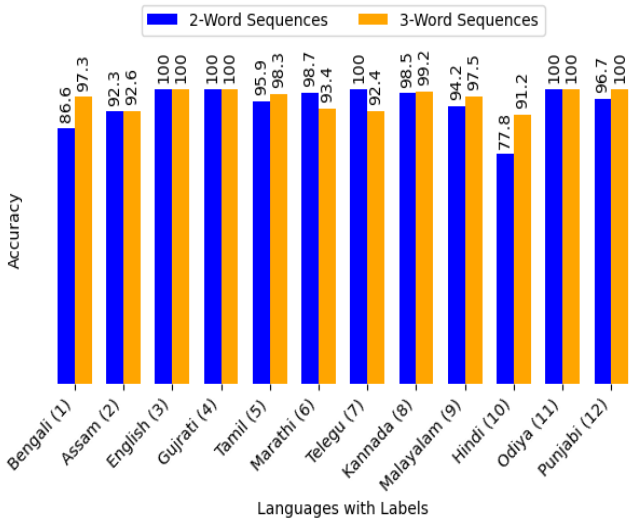


FIGURE 10. Comparison of Word Wise Accuracy across various Native Indian Languages. The x-axis indicates different language labels, and the y-axis represents accuracy. Blue bars denote accuracy for 2-word sequences, while orange bars represent accuracy for 3-word sequences.

2) XLM ROBERTa Base Native Character Wise confusion analysis

Individual testing sets for 3,4,5,6,7, and 8 characters were made. The results for each character are presented in Figure 11. The characters-wise confusion matrix for XLM Roberta is shown in Figure 12(a), Figure 12(b), Figure 12(c), and Figure 12(d). Upon examining the 5,6,7,8-char sequence confusion matrix distinct patterns emerged in testing accuracy. Marathi's testing accuracy displayed an increasing trend with the growth in the Number of characters, while Hindi words exhibited a decrease in testing accuracy, indicating misinterpretation of Hindi words. This decline is particularly noteworthy given the substantial word overlap between these two languages, as both languages share almost the same script. The intricate linguistic similarities posed challenges for accurate label predictions, further highlighting the complexities of language identification in this shared script context.

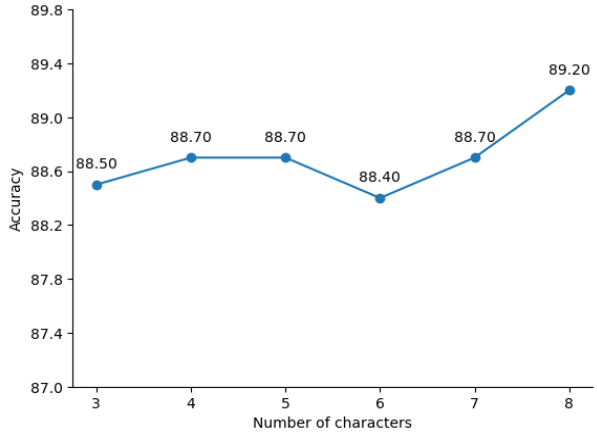
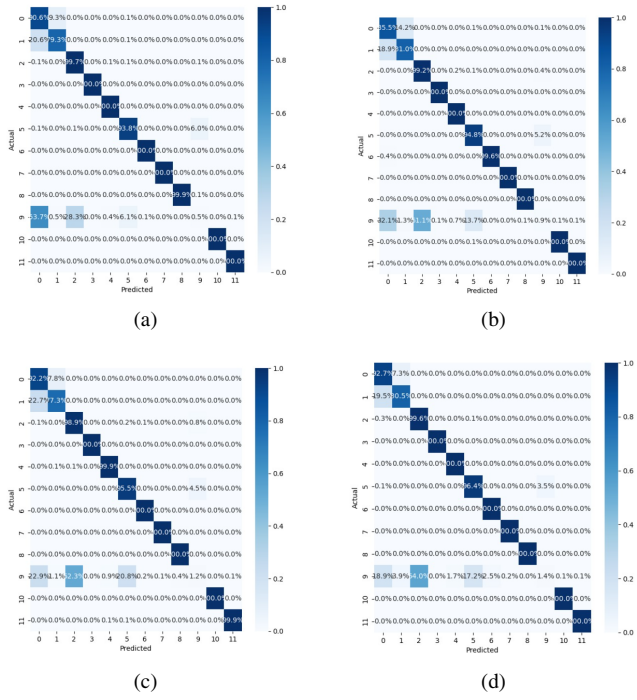


FIGURE 11. Relationship between the number of characters and their corresponding accuracy values. The figure provides insights into the performance metrics associated with different character lengths.



(a) (b) (c) (d)

FIGURE 12. (Characters-wise confusion matrices for the XLM-Roberta model. Subfigures (a), (b), (c), and (d) illustrate the confusion matrices for different character lengths, specifically, 5 characters, 6 characters, 7 characters, and 8 characters, providing a detailed comparison of model performance across varying character lengths.

3) XLM ROBERTa Romanized vs SVM Romanized Analysis

The confusion matrix and classification for both XLM Roberta Romanized and SVM Romanized are shown in Figure 13(a), 13(b),14(a) and 14(b). XLM Roberta Romanized performed better results compared to SVM Romanized. The

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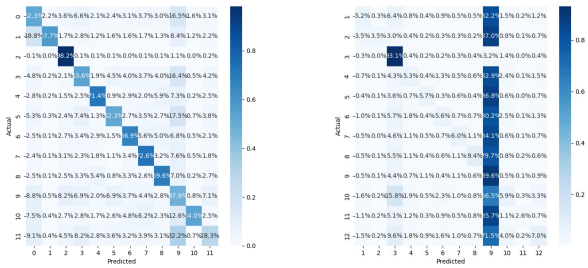
F1 score of XLM Roberta Romanized for all 12 Indian languages is higher than SVM Romanized.



(a) (b)

FIGURE 13. Confusion matrices for the Romanized testing sets. Subfigure (a) illustrates the confusion matrix for the XLM-Roberta Romanized testing set, while subfigure (b) showcases the confusion matrix for the SVM Romanized testing set

Label	Precision	Recall	f1-score	support
1	0.59	0.52	0.55	11276
2	0.84	0.58	0.68	3788
3	0.79	0.98	0.87	9999
4	0.65	0.54	0.59	12417
5	0.78	0.71	0.74	8824
6	0.66	0.52	0.58	7646
7	0.67	0.67	0.67	7681
8	0.64	0.73	0.68	7025
9	0.66	0.70	0.68	7613
10	0.21	0.48	0.30	6357
11	0.75	0.54	0.63	3093
12	0.50	0.28	0.36	8880
accuracy			0.61	94599
macro avg	0.64	0.60	0.61	94599
weighted avg	0.64	0.61	0.61	94599

(a)

Label	Precision	Recall	f1-score	support
1	0.44	0.05	0.09	11276
2	0.56	0.04	0.07	3788
3	0.64	0.93	0.76	9999
4	0.43	0.05	0.09	12417
5	0.55	0.06	0.10	8824
6	0.36	0.06	0.10	7646
7	0.44	0.06	0.11	7681
8	0.55	0.09	0.16	7025
9	0.10	0.90	0.18	7613
10	0.21	0.06	0.09	6357
11	0.45	0.03	0.05	3093
12	0.40	0.07	0.12	8880
accuracy			0.22	94599
macro avg	0.43	0.20	0.16	94599
weighted avg	0.43	0.22	0.18	94599

(b)

FIGURE 14. Classification reports for the Romanized testing sets, including (a) XLM-Roberta and (b) SVM. The columns present labels for different languages, including Precision, Recall, F1-score, and support. The table also includes total accuracy, macro-average, and weighted-average performance metrics.

C. ANALYSIS ON REAL-TIME DATA

Testing a language model on real-time datasets is a critical process that assesses its performance and adaptability to dynamic linguistic contexts. Real-time testing involves subjecting the model to diverse inputs, ranging from public grievance letters to administrative documents in different languages, often multilingual. This ensures it can effectively interpret and generate meaningful outputs across various domains. Our proposed pipeline has been tested on real-time data provided by the National Informatics Center (NIC). The data contained a total of 1600000 code-mixed sentences. The accuracy of our proposed pipeline is 92.76%.

VI. ERROR ANALYSIS

Overview of Error Analysis In language detection using the fine-tuned XLM-RoBERTa model, a notable challenge arises in differentiating closely related languages. Specifically, our model exhibited a higher confusion rate between Bengali and Assamese languages and between Hindi and Marathi as shown in Figure 15. This section delves into analyzing these errors, exploring the underlying reasons and their implications on the model's performance.

Bengali and Assamese: The confusion between Bengali and Assamese can be attributed to their linguistic similarities. Both languages share a significant number of common words and have similar grammatical structures stemming from their common origin in the Indo-Aryan language family. Furthermore, their distinct scripts bear a resemblance that can lead to misclassification by the model, especially in shorter texts where contextual cues are limited.

Hindi and Marathi: Similarly, the model's difficulty distinguishing between Hindi and Marathi arises from their close linguistic relationship as members of the Indo-Aryan language group. Hindi and Marathi have overlapping vocabularies and similar syntactic patterns. The scripts of both languages, Devanagari, are identical, adding another layer of complexity to the language detection task.

Impact on Model Performance The high degree of similarity in vocabulary and script between these language pairs poses a significant challenge for the model. In cases where the input text is short or lacks clear contextual indicators, the model is prone to misclassifying one language for the other. This is particularly evident in texts containing commonly used words or phrases prevalent in both languages.

Sentence	Target	Predicted
मुंबई रोल्टा कंपनीला भीषण आग	Marathi	Hindi
जें वैरियांही दिधलें	Marathi	Hindi
साधक साधना करिती	Marathi	Hindi
मैं उतना ही तैयार होता	Hindi	Marathi
क्या फ़ायदा होता	Hindi	Marathi
एकमात्र हैट्रिक	Hindi	Marathi
ঔষুধ কিনতে এলে	Bengali	Assamese
তাতেই চমক	Bengali	Assamese
ভাল লাগেনি	Bengali	Assamese
আমি মনে মনে বহি নাথাকো	Assamese	Bengali
সাইলাখ যেনিবা ভেকুলীবেং	Assamese	Bengali
অমানিশা হেমেন ডেকা	Assamese	Bengali

FIGURE 15. Error Analysis in Language Detection using Fine-Tuned XLM-Roberta Model. The "Target Language" column represents the true language of the sentence, and the "Predicted Language" column indicates the language predicted by the model. The model demonstrated a higher confusion rate specifically between Bengali and Assamese languages, as well as between Hindi and Marathi.

VII. LIMITATIONS OF THE STUDY

The XLM Roberta Native model attains the best accuracy as their training and testing data are mostly clean by removing noise like punctuation, special characters, and numbers.

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However, the XLM ROBERTa Base Romanized data from the Aksharantar and Abhijanam datasets might be noisy due to a mixed script. To solve this issue, better data is needed. This issue can degrade the performance of the XLM-ROBERTa Base Romanized model as there is a lot of confusion between all the 12 Indian languages.

VIII. CONCLUSION AND FUTURE SCOPE

This paper proposes a novel pipeline including XLM ROBERTa Native and XLM ROBERTa Romanized. There exist a few Indian Language identification baseline set for XLM ROBERTa Native and XLM ROBERTa Romanized Model. However, our proposed BERT-based models outperform other machine learning models. This is because of the following reasons. Bert's models made significant advancements in handling the out of vocabulary problem. These models use a sub-word tokenization scheme to convert the words into sub-words to produce the part of the unseen vocabulary. These models capture contextual information. This model uses Masked Language Modelling to learn the text, a form of unsupervised learning. This pipeline can be used for various Natural Language Processing Problems. As the results for XLM ROBERTa Romanized displayed, there can be a further possibility for improvement in the performance.

IX. ACKNOWLEDGMENT

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